VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH UNIVERSITY OF TECHNOLOGY

FACULTY OF CHEMICAL ENGINEERING

MECHANICAL PROCESSING

Laboratory Of Unit Operations (CH3015)

DOUBLE PIPE HEAT EXCHANGER

CLASS CC03 — GROUP 9 — SEMESTER 252

Instructor: Dr. Phạm Hoàng Huy Phước Lợi

Student's Name	Student's ID
Đinh Chấn Phong	2352902
Đinh Nguyễn Chính Nghĩa	2352804
Trần Hữu Đạt	2352245

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1 ABSTRACT

1.1 Purposes

- Get to know double-pipe heat exchanger, thermometers and flow meters.
- Determine heat transfer coefficient of heat exchanging between two hot and cold streams through a metal wall at different operation modes.
- Establish heat balance.

1.2 Methods

Study tube B and tube C at different operating modes, followed by calculating heat transfer coefficient between hot and cold fluid streams.

1.3 Results

Raw result: 1 $ft^3 = 0.02832\ m^3$

1. ***Tube B:*** Fluid flows vertically over the external surface of the inner tube. The two flows are perpendicular to each other.
2. ***Tube C:*** Simple tube, fluid flows longitudinally over external surface of the inner tube. The two flows are parallel to each other.

1.4 Result evaluation

- There are deviations between experimental data and theory due to errors from equipment and during the process of carry out the experiments.
- Heat transfer in tube type C is less efficient than in tube type B.
- Experimental heat transfer coefficient K_l is lower than overall heat transfer coefficient K_l^* because of heat loss and errors happening during experiment.

2 EXPERIMENTAL PRINCIPLE

Heat exchanging in a double-pipe heat exchanger is an example of complex heat transfer, where heat is exchanged between two fluid flows through a metal wall. This process involves:

- Convectional heat transfer from the hot stream to the wall.
- Conduction through the metal wall.
- Convectional heat transfer from the wall to the cold stream.

2.1 Heat Balance Equation

The heat balance equation for the two fluid streams is given by:

$$Q = G_1 C_{p1}(t_{1\text{in}} - t_{1\text{out}}) = G_2 C_{p2}(t_{2\text{out}} - t_{2\text{in}}) \quad [W] \tag{1}$$

Where:

- G_1, G_2: Hot and cold stream mass flow rate $[kg/s]$.
- C_{p1}, C_{p2}: Average specific heat capacity of hot and cold stream $[J/kg \cdot K]$.
- $t_{1\text{in}}, t_{1\text{out}}$: Inlet and outlet temperature of the hot stream $[°C]$.
- $t_{2\text{in}}, t_{2\text{out}}$: Inlet and outlet temperature of the cold stream $[°C]$.

2.2 Heat Transfer Equation

The heat transfer is also determined by:

$$Q = K_l \cdot \Delta t_{\log} \cdot L \tag{2}$$

Where:

- L: Effective tube length $[m]$.
- K_l: Experimental linear heat transfer coefficient $[W/m \cdot K]$.
- $\Delta t_{\log}$: Logarithmic Mean Temperature Difference (LMTD) $[K]$.

2.3 Logarithmic Mean Temperature Difference (LMTD)

The LMTD is defined as:

$$\Delta t_{\log} = \frac{\Delta t_{\max} - \Delta t_{\min}}{\ln\left(\frac{\Delta t_{\max}}{\Delta t_{\min}}\right)} \tag{3}$$

For counter-flow:

$$\Delta t_{\max} = t_{1\text{in}} - t_{2\text{out}}; \quad \Delta t_{\min} = t_{1\text{out}} - t_{2\text{in}} \tag{4}$$

2.4 Overall Heat Transfer Coefficient, K_l^*

The theoretical overall linear heat transfer coefficient K_l^* is calculated from the individual thermal transmittance coefficients of each stream and the wall conduction:

$$K_l^* = \frac{\pi}{\frac{1}{\alpha_1 d_1} + \frac{1}{2\lambda_w}\ln\left(\frac{d_2}{d_1}\right) + \frac{1}{\alpha_2 d_2}} \tag{5}$$

Where:

- α_1, α_2: Convective heat transfer coefficients $[W/m^2 \cdot K]$.
- λ_w: Thermal conductivity of the wall (copper) $[W/m \cdot K]$.
- d_1, d_2: Internal and external diameter of the inner tube $[m]$.

2.5 Thermal Transmittance Coefficient (α)

The coefficient α is determined from the Nusselt number (Nu):

$$\alpha = \frac{Nu \cdot \lambda}{l} \tag{6}$$

Where:

- λ: Thermal conductivity of the fluid $[W/m \cdot K]$.
- l: Characteristic dimension (e.g., equivalent diameter) $[m]$.

The Reynolds number (Re) determines the flow mode:

$$Re = \frac{w \cdot l}{\nu} \tag{7}$$

Where w is the flow velocity $[m/s]$ and ν is the kinematic viscosity $[m^2/s]$.

The equivalent diameter d_{eq} for non-circular sections (like the annulus in Tube C) is:

$$d_{\text{eq}} = \frac{4F}{\Pi} \tag{8}$$

Where F is the cross-sectional area and Π is the wetted circumference.

2.6 Dimensionless Numbers Correlation

The heat transfer coefficients are determined using dimensionless correlations specific to the flow geometry and regime.

2.6.1 Nusselt Number (Nu)

The Nusselt number represents the ratio of convective to conductive heat transfer.

Cross Flow (Tube B):

- $5 < Re < 10^3$: $Nu = 0.5 \cdot Re^{0.5} \cdot Pr^{0.38} \cdot \left(\frac{Pr}{Pr_w}\right)^{0.25} \cdot \epsilon_\psi$
- $10^3 < Re < 2 \cdot 10^5$: $Nu = 0.25 \cdot Re^{0.6} \cdot Pr^{0.38} \cdot \left(\frac{Pr}{Pr_w}\right)^{0.25} \cdot \epsilon_\psi$
- $2 \cdot 10^5 < Re < 2 \cdot 10^6$: $Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.37} \cdot \left(\frac{Pr}{Pr_w}\right)^{0.25} \cdot \epsilon_\psi$

Longitudinal Flow (Tube C):

- **Laminar Flow ($Re < 2320$):**

$$Nu = 0.15 \cdot Re^{0.33} \cdot Pr^{0.43} \cdot Gr^{0.1} \cdot \left(\frac{Pr}{Pr_w}\right)^{0.25} \cdot \epsilon_l \tag{9}$$

- **Transition Flow ($2320 < Re < 10000$):**

$$Nu = K \cdot Pr^{0.43} \cdot \left(\frac{Pr}{Pr_w}\right)^{0.25} \tag{10}$$

Where K depends on Re (e.g., $K = 2.2$ for $Re = 2200$, $K = 33$ for $Re = 10000$).

- **Turbulent Flow ($Re > 10^4$):**

$$Nu = 0.021 \cdot Re^{0.8} \cdot Pr^{0.43} \cdot \left(\frac{Pr}{Pr_w}\right)^{0.25} \cdot \epsilon_l \tag{11}$$

2.6.2 Prandtl Number (Pr)

The Prandtl number relates the momentum diffusivity to thermal diffusivity:

$$Pr = \frac{\nu}{a} = \frac{\mu \cdot C_p}{\lambda} \tag{12}$$

Where ν is kinematic viscosity and a is thermal diffusivity.

2.6.3 Grashof Number (Gr)

The Grashof number approximates the ratio of the buoyancy to viscous force acting on a fluid:

$$Gr = \frac{g \cdot \beta \cdot \Delta t \cdot l^3}{\nu^2} \tag{13}$$

Where:

- g: Gravitational acceleration ($9.81 m/s^2$).
- β: Volume expansion coefficient ($1/K$).
- Δt: Temperature difference between wall and fluid.

2.7 Temperature Distribution Diagram

The figure below illustrates the temperature gradient across the heat transfer surfaces.

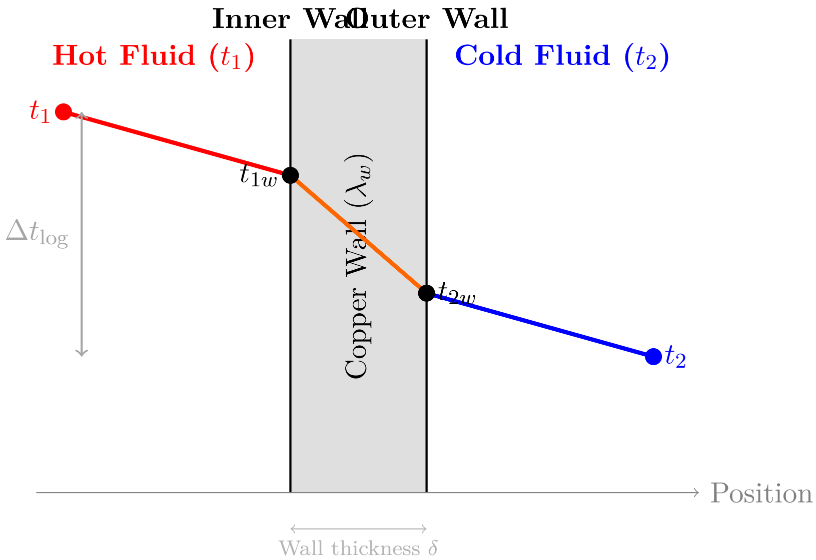


Figure 1: Temperature distribution across the double-pipe heat exchanger wall

2.8 Iterative Wall Temperature Calculation

If the wall temperatures t_{1w}, t_{2w} are not directly measured, they are determined iteratively using:

$$\Delta t_1 = t_1 - t_{1w} = \frac{Q}{\alpha_1 \cdot F_1} \tag{14}$$

$$\Delta t_2 = t_{2w} - t_2 = \frac{Q}{\alpha_2 \cdot F_2} \tag{15}$$

The procedure is repeated until the error between assumed and calculated Δt is less than 5%.

3 EXPERIMENT EQUIPMENT AND METHOD

3.1 Experiment equipment

There are 3 types of heat transfer surfaces:

- **Type A:** Elastic tube.
- **Type B:** Fluid flows vertically over the external surface of the inner tube. The two flows are perpendicular to each other.
- **Type C:** Simple tube, fluid flows longitudinally over the external surface of the inner tube. The two flows are parallel to each other.

Table 1: Tube dimensions

Tube	**Diameter** (mm)		**Length** (mm)
	Inner tube	**Outer tube**	
B	$\Phi 14/16$	$\Phi 26/28$	925
C	$\Phi 14/16$	$\Phi 26/28$	1000

3.2 Method

- Supply water to heater, turn on resistors to heat up the water.
- Study flows of the hot and cold streams and how to adjust valves.
- Adjust flow rate of the hot streams using valve 3 (reflux valve) respectively at the flow rate of (L/min): 4, 6, 8, 10, 12.
- For each flow rate of the hot stream, adjust flow rate of the cold stream using valve 4 respectively at the flow rate of (L/min): 4, 6, 8, 10, 12.
- Drive valve II to measure the flow rate of the hot and cold stream.
- Measure inlet and outlet temperature of the cold stream ($t_{2\text{in}}, t_{2\text{out}}$), and the hot stream ($t_{1\text{in}}, t_{1\text{out}}$) in each experiment mode when it is stable. The hot steam is measured and refluxed continuously to avoid heat loss.
- In this experiment, the streams are counter-flow.

4 RESULTS

4.1 Raw Experimental Data

Table 2: Tube B Experimental Data (Flow rates in L/min, Temperatures in °C)

Cold Flow	**Hot: 4**				**Hot: 6**				**Hot: 8**				**Hot: 10**				**Hot: 12**			
(L/min)	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$
4	54	49	34	40	59	54	36	42	64	58	39	45	69	63	40	47	71	66	41	48
6	56	49	35	40	60	54	36	42	65	58	38	44	70	63	40	46	71	66	41	47
8	56	50	34	40	60	54	36	42	66	58	37	43	70	63	39	44	71	65	40	46
10	57	50	34	40	62	54	36	41	67	58	37	42	70	63	39	44	72	65	40	45
12	58	51	34	40	63	55	36	41	68	59	37	42	71	63	39	44	72	66	40	45

Table 3: Tube C_2 Experimental Data (Flow rates in L/min, Temperatures in °C)

Cold Flow	**Hot: 4**				**Hot: 6**				**Hot: 8**				**Hot: 10**				**Hot: 12**			
(L/min)	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$	$T_{1,in}$	$T_{1,out}$	$T_{2,in}$	$T_{2,out}$
4	61	50	32	36	67	56	35	39	69	59	37	41	69	61	39	43	71	63	40	44
6	63	52	32	35	67	56	34	38	69	59	36	40	70	61	38	42	71	63	40	43
8	65	53	32	35	67	56	34	37	70	59	36	40	71	61	38	41	71	62	39	42
10	66	53	32	35	68	56	34	37	69	59	36	39	71	61	38	41	72	62	39	41
12	66	53	33	35	69	56	34	37	70	59	36	39	71	61	37	40	72	62	38	41

4.2 Calculated Experimental Results

4.2.1 Fluid Physical Parameters

The physical properties of water are interpolated based on the average flow temperature $T_{avg} = (T_{in} + T_{out})/2$.

Table 4: Fluid Parameters - Hot stream - Tube B

No.	G_1 (L/min)	G_2 (L/min)	$T_{1,in}$ (°C)	$T_{1,out}$ (°C)	$T_{1,avg}$ (°C)	ρ_1 (kg/m^3)	$C_{p,1}$ (J/kg·K)	λ_1 (W/m·K)	μ_1 (Pa·s)	Pr_1
1	4	4	54	49	51.50	987.37	4182	0.6455	5.35e-04	3.46
2	4	6	56	49	52.50	986.88	4182	0.6465	5.27e-04	3.41
3	4	8	56	50	53.00	986.63	4182	0.6470	5.23e-04	3.38
4	4	10	57	50	53.50	986.38	4182	0.6475	5.19e-04	3.35
5	4	12	58	51	54.50	985.89	4183	0.6485	5.11e-04	3.29
6	6	4	59	54	56.50	984.92	4184	0.6505	4.95e-04	3.18
7	6	6	60	54	57.00	984.67	4184	0.6510	4.91e-04	3.15
8	6	8	60	54	57.00	984.67	4184	0.6510	4.91e-04	3.15
9	6	10	62	54	58.00	984.18	4184	0.6520	4.83e-04	3.09
10	6	12	63	55	59.00	983.69	4185	0.6530	4.75e-04	3.04
11	8	4	64	58	61.00	982.66	4186	0.6549	4.61e-04	2.94
12	8	6	65	58	61.50	982.39	4186	0.6553	4.58e-04	2.92
13	8	8	66	58	62.00	982.12	4186	0.6558	4.54e-04	2.89
14	8	10	67	58	62.50	981.85	4186	0.6562	4.51e-04	2.87
15	8	12	68	59	63.50	981.31	4187	0.6572	4.45e-04	2.83
16	10	4	69	63	66.00	979.96	4188	0.6594	4.29e-04	2.72
17	10	6	70	63	66.50	979.69	4188	0.6599	4.26e-04	2.70
18	10	8	70	63	66.50	979.69	4188	0.6599	4.26e-04	2.70
19	10	10	70	63	66.50	979.69	4188	0.6599	4.26e-04	2.70
20	10	12	71	63	67.00	979.42	4188	0.6603	4.23e-04	2.68
21	12	4	71	66	68.50	978.61	4189	0.6617	4.13e-04	2.61
22	12	6	71	66	68.50	978.61	4189	0.6617	4.13e-04	2.61
23	12	8	71	65	68.00	978.88	4189	0.6612	4.17e-04	2.64
24	12	10	72	65	68.50	978.61	4189	0.6617	4.13e-04	2.61
25	12	12	72	66	69.00	978.34	4190	0.6621	4.10e-04	2.59

Table 5: Fluid Parameters - Cold stream - Tube B

No.	G_1 (L/min)	G_2 (L/min)	$T_{2,in}$ (°C)	$T_{2,out}$ (°C)	$T_{2,avg}$ (°C)	ρ_2 (kg/m^3)	$C_{p,2}$ (J/kg·K)	λ_2 (W/m·K)	μ_2 (Pa·s)	Pr_2
1	4	4	34	40	37.00	993.25	4179	0.6262	6.97e-04	4.65
2	4	6	35	40	37.50	993.08	4179	0.6270	6.89e-04	4.60
3	4	8	34	40	37.00	993.25	4179	0.6262	6.97e-04	4.65
4	4	10	34	40	37.00	993.25	4179	0.6262	6.97e-04	4.65
5	4	12	34	40	37.00	993.25	4179	0.6262	6.97e-04	4.65
6	6	4	36	42	39.00	992.55	4179	0.6294	6.68e-04	4.43
7	6	6	36	42	39.00	992.55	4179	0.6294	6.68e-04	4.43
8	6	8	36	42	39.00	992.55	4179	0.6294	6.68e-04	4.43
9	6	10	36	41	38.50	992.73	4179	0.6286	6.75e-04	4.49

Table 5 – continued from previous page

No.	G_1 (L/min)	G_2 (L/min)	$T_{2,in}$ (°C)	$T_{2,out}$ (°C)	$T_{2,avg}$ (°C)	ρ_2 (kg/m^3)	$C_{p,2}$ (J/kg·K)	λ_2 (W/m·K)	μ_2 (Pa·s)	Pr_2
10	6	12	36	41	38.50	992.73	4179	0.6286	6.75e-04	4.49
11	8	4	39	45	42.00	991.38	4179	0.6336	6.32e-04	4.17
12	8	6	38	44	41.00	991.79	4179	0.6323	6.42e-04	4.24
13	8	8	37	43	40.00	992.20	4179	0.6310	6.53e-04	4.32
14	8	10	37	42	39.50	992.38	4179	0.6302	6.60e-04	4.38
15	8	12	37	42	39.50	992.38	4179	0.6302	6.60e-04	4.38
16	10	4	40	47	43.50	990.76	4180	0.6356	6.16e-04	4.05
17	10	6	40	46	43.00	990.97	4180	0.6349	6.21e-04	4.09
18	10	8	39	44	41.50	991.59	4179	0.6330	6.37e-04	4.20
19	10	10	39	44	41.50	991.59	4179	0.6330	6.37e-04	4.20
20	10	12	39	44	41.50	991.59	4179	0.6330	6.37e-04	4.20
21	12	4	41	48	44.50	990.36	4180	0.6369	6.05e-04	3.97
22	12	6	41	47	44.00	990.56	4180	0.6362	6.11e-04	4.01
23	12	8	40	46	43.00	990.97	4180	0.6349	6.21e-04	4.09
24	12	10	40	45	42.50	991.18	4180	0.6342	6.26e-04	4.13
25	12	12	40	45	42.50	991.18	4180	0.6342	6.26e-04	4.13

Table 6: Fluid Parameters - Hot stream - Tube C

No.	G_1 (L/min)	G_2 (L/min)	$T_{1,in}$ (°C)	$T_{1,out}$ (°C)	$T_{1,avg}$ (°C)	ρ_1 (kg/m^3)	$C_{p,1}$ (J/kg·K)	λ_1 (W/m·K)	μ_1 (Pa·s)	Pr_1
1	4	4	61	50	55.50	985.41	4183	0.6495	5.03e-04	3.24
2	4	6	63	52	57.50	984.43	4184	0.6515	4.87e-04	3.12
3	4	8	65	53	59.00	983.69	4185	0.6530	4.75e-04	3.04
4	4	10	66	53	59.50	983.45	4185	0.6535	4.71e-04	3.01
5	4	12	66	53	59.50	983.45	4185	0.6535	4.71e-04	3.01
6	6	4	67	56	61.50	982.39	4186	0.6553	4.58e-04	2.92
7	6	6	67	56	61.50	982.39	4186	0.6553	4.58e-04	2.92
8	6	8	67	56	61.50	982.39	4186	0.6553	4.58e-04	2.92
9	6	10	68	56	62.00	982.12	4186	0.6558	4.54e-04	2.89
10	6	12	69	56	62.50	981.85	4186	0.6562	4.51e-04	2.87
11	8	4	69	59	64.00	981.04	4187	0.6576	4.42e-04	2.81
12	8	6	69	59	64.00	981.04	4187	0.6576	4.42e-04	2.81
13	8	8	70	59	64.50	980.77	4187	0.6581	4.39e-04	2.79
14	8	10	69	59	64.00	981.04	4187	0.6576	4.42e-04	2.81
15	8	12	70	59	64.50	980.77	4187	0.6581	4.39e-04	2.79
16	10	4	69	61	65.00	980.50	4188	0.6585	4.36e-04	2.76
17	10	6	70	61	65.50	980.23	4188	0.6590	4.32e-04	2.74
18	10	8	71	61	66.00	979.96	4188	0.6594	4.29e-04	2.72
19	10	10	71	61	66.00	979.96	4188	0.6594	4.29e-04	2.72
20	10	12	71	61	66.00	979.96	4188	0.6594	4.29e-04	2.72
21	12	4	71	63	67.00	979.42	4188	0.6603	4.23e-04	2.68
22	12	6	71	63	67.00	979.42	4188	0.6603	4.23e-04	2.68
23	12	8	71	62	66.50	979.69	4188	0.6599	4.26e-04	2.70
24	12	10	72	62	67.00	979.42	4188	0.6603	4.23e-04	2.68
25	12	12	72	62	67.00	979.42	4188	0.6603	4.23e-04	2.68

Table 7: Fluid Parameters - Cold stream - Tube C

No.	G_1 (L/min)	G_2 (L/min)	$T_{2,in}$ (°C)	$T_{2,out}$ (°C)	$T_{2,avg}$ (°C)	ρ_2 (kg/m^3)	$C_{p,2}$ (J/kg·K)	λ_2 (W/m·K)	μ_2 (Pa·s)	Pr_2
1	4	4	32	36	34.00	994.30	4178	0.6214	7.40e-04	4.98
2	4	6	32	35	33.50	994.48	4178	0.6206	7.47e-04	5.04
3	4	8	32	35	33.50	994.48	4178	0.6206	7.47e-04	5.04
4	4	10	32	35	33.50	994.48	4178	0.6206	7.47e-04	5.04
5	4	12	33	35	34.00	994.30	4178	0.6214	7.40e-04	4.98
6	6	4	35	39	37.00	993.25	4179	0.6262	6.97e-04	4.65
7	6	6	34	38	36.00	993.60	4179	0.6246	7.11e-04	4.76
8	6	8	34	37	35.50	993.78	4179	0.6238	7.18e-04	4.82
9	6	10	34	37	35.50	993.78	4179	0.6238	7.18e-04	4.82
10	6	12	34	37	35.50	993.78	4179	0.6238	7.18e-04	4.82
11	8	4	37	41	39.00	992.55	4179	0.6294	6.68e-04	4.43
12	8	6	36	40	38.00	992.90	4179	0.6278	6.82e-04	4.54
13	8	8	36	40	38.00	992.90	4179	0.6278	6.82e-04	4.54
14	8	10	36	39	37.50	993.08	4179	0.6270	6.89e-04	4.60
15	8	12	36	39	37.50	993.08	4179	0.6270	6.89e-04	4.60
16	10	4	39	43	41.00	991.79	4179	0.6323	6.42e-04	4.24
17	10	6	38	42	40.00	992.20	4179	0.6310	6.53e-04	4.32
18	10	8	38	41	39.50	992.38	4179	0.6302	6.60e-04	4.38
19	10	10	38	41	39.50	992.38	4179	0.6302	6.60e-04	4.38
20	10	12	37	40	38.50	992.73	4179	0.6286	6.75e-04	4.49
21	12	4	40	44	42.00	991.38	4179	0.6336	6.32e-04	4.17
22	12	6	40	43	41.50	991.59	4179	0.6330	6.37e-04	4.20
23	12	8	39	42	40.50	992.00	4179	0.6317	6.48e-04	4.28
24	12	10	39	41	40.00	992.20	4179	0.6310	6.53e-04	4.32
25	12	12	38	41	39.50	992.38	4179	0.6302	6.60e-04	4.38

4.2.2 Heat Balance and Loss

The heat transfer rate for the hot stream (Q_1) and cold stream (Q_2) are calculated as:

$$Q = \dot{m}C_p(T_{in} - T_{out}) = \rho G_{L/min}\frac{10^{-3}}{60}C_p\Delta T \tag{16}$$

Sample Calculation (Row 1, Tube B):

- $Q_1 = 987.37 \times 4 \times \frac{10^{-3}}{60} \times 4182 \times (54 - 49) = 1376.26\,\mathrm{W}$
- $Q_2 = 993.25 \times 4 \times \frac{10^{-3}}{60} \times 4179 \times (40 - 34) = 1660.20\,\mathrm{W}$
- $Q_{loss} = Q_1 - Q_2 = 1376.26 - 1660.20 = -283.94\,\mathrm{W}$

Table 8: Heat balance - Tube B

No.	ΔT_1 (°C)	Q_1 (W)	ΔT_2 (°C)	Q_2 (W)	Q_{loss} (W)
1	5	1376.26	6	1660.20	-283.94

Table 8 – continued from previous page

No.	ΔT_1 (°C)	Q_1 (W)	ΔT_2 (°C)	Q_2 (W)	Q_{loss} (W)
2	7	1925.99	5	2074.91	-148.92
3	6	1650.51	6	3320.40	-1669.88
4	7	1925.21	6	4150.49	-2225.28
5	7	1924.44	6	4980.59	-3056.15
6	5	2060.25	6	1659.11	401.14
7	6	2471.80	6	2488.66	-16.86
8	6	2471.80	6	3318.21	-846.42
9	8	3294.40	5	3457.04	-162.64
10	8	3293.08	5	4148.45	-855.37
11	6	3290.34	6	1657.35	1632.99
12	7	3837.90	6	2486.93	1350.97
13	8	4385.23	6	3317.12	1068.11
14	9	4932.32	5	3455.90	1476.42
15	9	4930.20	5	4147.09	783.11
16	6	4104.07	7	1932.51	2171.56
17	7	4787.05	6	2485.11	2301.94
18	7	4787.05	5	2762.75	2024.30
19	7	4787.05	5	3453.44	1333.61
20	8	5469.73	5	4144.13	1325.60
21	5	4099.64	7	1931.81	2167.84
22	5	4099.64	6	2484.21	1615.44
23	6	4920.63	6	3313.49	1607.15
24	7	5739.50	5	3452.18	2287.32
25	6	4918.51	5	4142.62	775.89

Table 9: Heat balance - Tube C

No.	ΔT_1 (°C)	Q_1 (W)	ΔT_2 (°C)	Q_2 (W)	Q_{loss} (W)
1	11	3022.91	4	1107.89	1915.02
2	11	3020.48	3	1246.58	1773.90
3	12	3293.08	3	1662.11	1630.97
4	13	3566.78	3	2077.63	1489.15
5	13	3566.78	2	1661.83	1904.95
6	11	4523.24	4	1106.80	3416.44
7	11	4523.24	4	1660.74	2862.50
8	11	4523.24	3	1661.02	2862.23
9	12	4933.39	3	2076.27	2857.12
10	13	5343.35	3	2491.52	2851.83
11	10	5476.82	4	1106.07	4370.75
12	10	5476.82	4	1659.65	3817.17
13	11	6023.20	4	2212.87	3810.33
14	10	5476.82	3	2074.91	3401.91
15	11	6023.20	3	2489.89	3533.32
16	8	5474.46	4	1105.30	4369.15
17	9	6157.44	4	1658.56	4498.88
18	10	6840.12	3	1658.83	5181.29
19	10	6840.12	3	2073.54	4766.58

Table 9 – continued from previous page

No.	ΔT_1 (°C)	Q_1 (W)	ΔT_2 (°C)	Q_2 (W)	Q_{loss} (W)
20	10	6840.12	3	2489.07	4351.05
21	8	6563.68	4	1104.90	5458.78
22	8	6563.68	3	1243.24	5320.44
23	9	7385.74	3	1658.26	5727.48
24	10	8204.60	2	1382.13	6822.47
25	10	8204.60	3	2488.25	5716.35

For counter-current flow, the LMTD is calculated as:

$$\Delta T_{LMTD} = \frac{\Delta T_{max} - \Delta T_{min}}{\ln(\frac{\Delta T_{max}}{\Delta T_{min}})} \tag{17}$$

where $\Delta T_{max} = T_{1,in} - T_{2,out}$ and $\Delta T_{min} = T_{1,out} - T_{2,in}$ (for pure counter-flow).

Sample Calculation (Row 1, Tube B):

- $\Delta T_{max} = 54 - 40 = 14°\text{C}$
- $\Delta T_{min} = 49 - 34 = 15°\text{C}$
- $\Delta T_{LMTD} = \frac{15-14}{\ln(15/14)} = 14.49°\text{C}$

Table 10: LMTD Analysis - Tube B

No.	ΔT_{max} (°C)	ΔT_{min} (°C)	ΔT_{LMTD} (°C)
1	14.00	15.00	14.49
2	16.00	14.00	14.98
3	16.00	16.00	16.00
4	17.00	16.00	16.49
5	18.00	17.00	17.50
6	17.00	18.00	17.50
7	18.00	18.00	18.00
8	18.00	18.00	18.00
9	21.00	18.00	19.46
10	22.00	19.00	20.46
11	19.00	19.00	19.00
12	21.00	20.00	20.50
13	23.00	21.00	21.98
14	25.00	21.00	22.94
15	26.00	22.00	23.94
16	22.00	23.00	22.50
17	24.00	23.00	23.50
18	26.00	24.00	24.99
19	26.00	24.00	24.99
20	27.00	24.00	25.47

Table 10 – continued from previous page

No.	ΔT_{max} (°C)	ΔT_{min} (°C)	ΔT_{LMTD} (°C)
21	23.00	25.00	23.99
22	24.00	25.00	24.50
23	25.00	25.00	25.00
24	27.00	25.00	25.99
25	27.00	26.00	26.50

Table 11: LMTD Analysis - Tube C

No.	ΔT_{max} (°C)	ΔT_{min} (°C)	ΔT_{LMTD} (°C)
1	25.00	18.00	21.31
2	28.00	20.00	23.78
3	30.00	21.00	25.23
4	31.00	21.00	25.68
5	31.00	20.00	25.10
6	28.00	21.00	24.33
7	29.00	22.00	25.34
8	30.00	22.00	25.79
9	31.00	22.00	26.24
10	32.00	22.00	26.69
11	28.00	22.00	24.88
12	29.00	23.00	25.88
13	30.00	23.00	26.35
14	30.00	23.00	26.35
15	31.00	23.00	26.80
16	26.00	22.00	23.94
17	28.00	23.00	25.42
18	30.00	23.00	26.35
19	30.00	23.00	26.35
20	31.00	24.00	27.35
21	27.00	23.00	24.95
22	28.00	23.00	25.42
23	29.00	23.00	25.88
24	31.00	23.00	26.80
25	31.00	24.00	27.35

4.2.3 Flow Regimes and Correlations

The flow velocity (w) and Reynolds number (Re) are determined to choose the appropriate Nusselt correlation (Nu).

Sample Calculation (Row 1, Tube B):

- $w_1 = \frac{4\times10^{-3}/60}{\frac{\pi}{4}\times0.014^2} = 0.4331\,\text{m/s}$
- $Re_1 = \frac{987.37\times0.4331\times0.014}{5.35\times10^{-4}} = 11189.6$ (Turbulent)

- $Nu_1 = 0.021 \times Re_1^{0.8} \times Pr_1^{0.43} = 0.021 \times 11189.6^{0.8} \times 3.46^{0.43} = 114.90$
- $\alpha_1 = Nu_1 \times \lambda_1 / d_{in} = 114.90 \times 0.6455/0.014 = 5297.53\,\mathrm{W/m^2 \cdot K}$

Table 12: Flow Mode analysis - Tube B

No.	$\boldsymbol{w_1}$ (m/s)	$\boldsymbol{Re_1}$	$\boldsymbol{\lambda_1}$ (W/m·K)	$\boldsymbol{w_2}$ (m/s)	$\boldsymbol{Re_2}$	$\boldsymbol{\lambda_2}$ (W/m·K)
1	0.4331	11189.6	0.6455	0.2021	2882.1	0.6262
2	0.4331	11353.8	0.6465	0.3032	4367.8	0.6270
3	0.4331	11437.8	0.6470	0.4042	5764.2	0.6262
4	0.4331	11523.1	0.6475	0.5053	7205.2	0.6262
5	0.4331	11697.7	0.6485	0.6063	8646.3	0.6262
6	0.6496	18095.7	0.6505	0.2021	3005.2	0.6294
7	0.6496	18238.6	0.6510	0.3032	4507.8	0.6294
8	0.6496	18238.6	0.6510	0.4042	6010.4	0.6294
9	0.6496	18531.5	0.6520	0.5053	7433.5	0.6286
10	0.6496	18834.2	0.6530	0.6063	8920.2	0.6286
11	0.8661	25864.6	0.6549	0.2021	3171.2	0.6336
12	0.8661	26035.5	0.6553	0.3032	4680.3	0.6323
13	0.8661	26208.8	0.6558	0.4042	6141.7	0.6310
14	0.8661	26384.5	0.6562	0.5053	7594.1	0.6302
15	0.8661	26743.4	0.6572	0.6063	9112.9	0.6302
16	1.0827	34608.2	0.6594	0.2021	3251.1	0.6356
17	1.0827	34854.5	0.6599	0.3032	4836.0	0.6349
18	1.0827	34854.5	0.6599	0.4042	6291.0	0.6330
19	1.0827	34854.5	0.6599	0.5053	7863.8	0.6330
20	1.0827	35104.4	0.6603	0.6063	9436.5	0.6330
21	1.2992	43052.5	0.6617	0.2021	3306.7	0.6369
22	1.2992	43052.5	0.6617	0.3032	4918.0	0.6362
23	1.2992	42738.8	0.6612	0.4042	6448.1	0.6349
24	1.2992	43052.5	0.6617	0.5053	7993.5	0.6342
25	1.2992	43371.1	0.6621	0.6063	9592.2	0.6342

Table 13: Flow Mode analysis - Tube C

No.	$\boldsymbol{w_1}$ (m/s)	$\boldsymbol{Re_1}$	$\boldsymbol{\lambda_1}$ (W/m·K)	$\boldsymbol{w_2}$ (m/s)	$\boldsymbol{Re_2}$	$\boldsymbol{\lambda_2}$ (W/m·K)
1	0.4331	11877.8	0.6495	0.2021	2715.5	0.6214
2	0.4331	12255.9	0.6515	0.3032	4034.5	0.6206
3	0.4331	12556.1	0.6530	0.4042	5379.3	0.6206
4	0.4331	12659.6	0.6535	0.5053	6724.2	0.6206
5	0.4331	12659.6	0.6535	0.6063	8146.6	0.6214
6	0.6496	19526.6	0.6553	0.2021	2882.1	0.6262
7	0.6496	19526.6	0.6553	0.3032	4236.5	0.6246
8	0.6496	19526.6	0.6553	0.4042	5592.6	0.6238
9	0.6496	19656.6	0.6558	0.5053	6990.7	0.6238
10	0.6496	19788.4	0.6562	0.6063	8388.9	0.6238
11	0.8661	26926.6	0.6576	0.2021	3005.2	0.6294
12	0.8661	26926.6	0.6576	0.3032	4413.5	0.6278

Table 13 – continued from previous page

No.	w_1 (m/s)	Re_1	λ_1 (W/m·K)	w_2 (m/s)	Re_2	λ_2 (W/m·K)
13	0.8661	27112.5	0.6581	0.4042	5884.7	0.6278
14	0.8661	26926.6	0.6576	0.5053	7279.7	0.6270
15	0.8661	27112.5	0.6581	0.6063	8735.7	0.6270
16	1.0827	34126.4	0.6585	0.2021	3120.2	0.6323
17	1.0827	34365.6	0.6590	0.3032	4606.2	0.6310
18	1.0827	34608.2	0.6594	0.4042	6075.3	0.6302
19	1.0827	34608.2	0.6594	0.5053	7594.1	0.6302
20	1.0827	34608.2	0.6594	0.6063	8920.2	0.6286
21	1.2992	42125.3	0.6603	0.2021	3171.2	0.6336
22	1.2992	42125.3	0.6603	0.3032	4718.3	0.6330
23	1.2992	41825.4	0.6599	0.4042	6190.6	0.6317
24	1.2992	42125.3	0.6603	0.5053	7677.1	0.6310

Table 14: Heat transfer coefficients - Tube B

No.	Pr_1	Nu_1	α_1 (W/m^2·K)	Nu_2	α_2 (W/m^2·K)
1	3.46	114.90	5297.53	9.90	620.09
2	3.41	109.52	5057.45	21.93	1374.72
3	3.38	107.50	4968.06	33.39	2091.08
4	3.35	106.27	4914.94	44.96	2815.65
5	3.29	105.14	4870.06	56.38	3530.70
6	3.18	148.81	6914.53	10.91	686.39
7	3.15	142.14	6609.58	23.12	1455.48
8	3.15	139.05	6466.03	35.05	2205.74
9	3.09	137.28	6393.27	46.69	2935.15
10	3.04	136.02	6344.17	58.53	3679.07
11	2.94	179.57	8400.24	12.15	769.82
12	2.92	171.98	8050.51	24.50	1549.12
13	2.89	168.26	7881.78	36.48	2301.87
14	2.87	165.66	7765.16	48.30	3043.77
15	2.83	163.84	7690.31	60.31	3800.83
16	2.72	203.86	9601.74	12.60	800.78
17	2.70	199.24	9390.43	25.69	1630.92
18	2.70	193.39	9114.69	37.47	2371.95
19	2.70	190.72	8989.26	49.82	3153.35
20	2.68	189.41	8933.38	62.30	3942.99
21	2.61	226.57	10707.70	12.90	821.75
22	2.61	221.44	10465.49	26.02	1655.26
23	2.64	217.60	10276.74	38.46	2442.02
24	2.61	214.45	10134.99	50.80	3221.69
25	2.59	211.92	10022.10	63.14	4004.76

Table 15: heat transfer coefficients - Tube C

No.	$\boldsymbol{Pr_1}$	$\boldsymbol{Nu_1}$	$\boldsymbol{\alpha_1}$ (W/m^2·K)	$\boldsymbol{Nu_2}$	$\boldsymbol{\alpha_2}$ (W/m^2·K)
1	3.24	68.07	3158.03	9.05	562.08
2	3.12	62.97	2930.13	20.19	1253.15
3	3.04	61.61	2873.48	31.48	1953.76
4	3.01	60.59	2828.32	42.50	2637.44
5	3.01	60.04	2802.54	53.80	3343.26
6	2.92	96.52	4518.28	10.39	650.51
7	2.92	90.52	4237.47	21.97	1372.21
8	2.92	87.47	4094.39	33.16	2068.77
9	2.89	86.25	4040.23	44.66	2785.67
10	2.87	85.33	4000.07	56.00	3493.43
11	2.81	121.67	5714.92	11.17	702.75
12	2.81	116.37	5466.12	23.25	1459.90
13	2.79	113.34	5327.30	35.25	2212.95
14	2.81	110.54	5192.41	46.43	2911.31
15	2.79	109.54	5148.97	58.16	3646.53
16	2.76	145.53	6845.01	11.85	749.50
17	2.74	141.05	6639.04	24.44	1542.01
18	2.72	137.00	6452.79	36.34	2289.98
19	2.72	134.69	6343.78	48.33	3045.77
20	2.72	132.76	6253.08	59.49	3739.24
21	2.68	168.57	7950.43	12.15	769.82
22	2.68	164.30	7749.09	25.09	1587.83
23	2.70	159.90	7536.19	37.08	2341.95
24	2.68	157.33	7420.21	49.04	3094.16
25	2.68	155.58	7337.78	60.80	3831.72

4.2.4 Overall Heat Transfer Coefficient and K-Value Analysis

The experimental heat transfer coefficient (K_{exp}) is calculated as:

$$K_{exp} = \frac{Q_1}{L \cdot \Delta T_{LMTD}} \tag{18}$$

The theoretical coefficient (K^*) is determined by iterating the wall temperature T_w until:

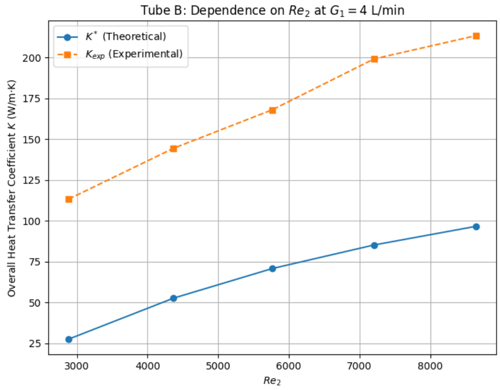
$$q = \alpha_1(T_1 - T_w) = \frac{\lambda_{wall}}{\delta}(T_w - T_{w,out}) = \dots \tag{19}$$

Sample Calculation (Row 1, Tube B):

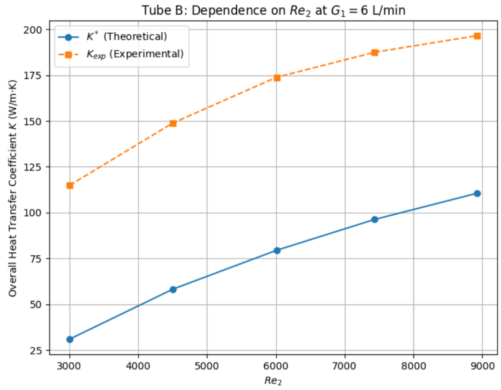
- $K_{exp} = \frac{1376.26}{0.84 \cdot 14.49} = 113.24\,\mathrm{W/m \cdot K}$
- $K^* = \left(\frac{1}{\alpha_1 d_1} + \frac{1}{2\lambda_w} \ln \frac{d_2}{d_1} + \frac{1}{\alpha_2 d_2}\right)^{-1} \cdot \pi = 27.45\,\mathrm{W/m \cdot K}$
- Error (%) $= \frac{|113.24 - 27.45|}{113.24} \times 100 = 75.76\%$

Table 16: K stars and K experimental - Tube B

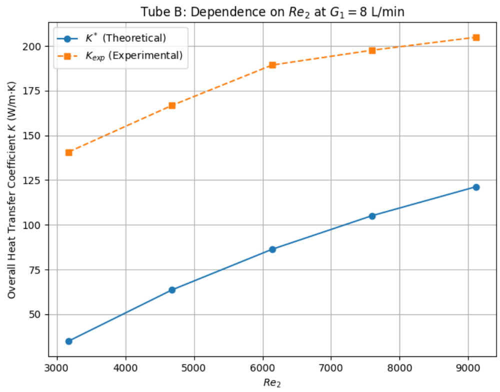
No.	$\boldsymbol{K^*}$ (W/m·K)	$\boldsymbol{K_{exp}}$ (W/m·K)	$\boldsymbol{Error}$ (%)
1	27.45	113.24	75.76
2	52.57	144.39	63.59
3	70.69	167.94	57.90
4	85.13	199.10	57.24
5	96.54	213.34	54.75
6	30.93	114.91	73.08
7	58.26	148.96	60.89
8	79.42	173.87	54.32
9	96.25	187.52	48.67
10	110.54	196.57	43.76
11	34.96	140.76	75.16
12	63.61	166.81	61.87
13	86.34	189.38	54.41
14	105.05	197.64	46.85
15	121.27	204.92	40.82
16	36.67	145.05	74.72
17	68.14	167.30	59.27
18	91.43	163.33	44.02
19	112.44	178.27	36.93
20	130.79	204.03	35.90
21	37.90	135.92	72.12
22	70.19	145.28	51.68
23	96.02	178.04	46.07
24	118.01	191.19	38.27
25	137.15	184.85	25.81



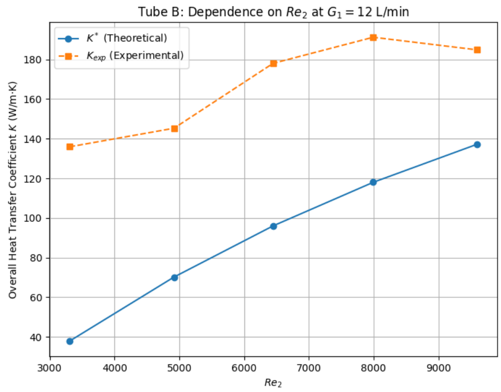
(a) $G_1 = 4$ L/min



(b) $G_1 = 6$ L/min

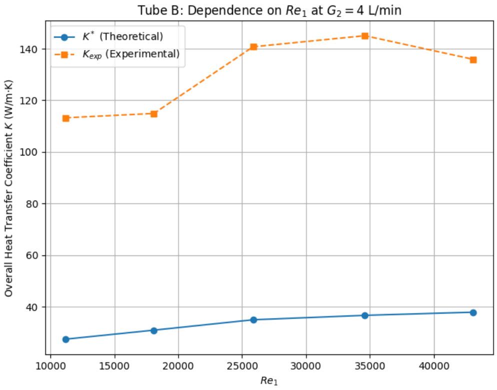


(c) $G_1 = 8$ L/min

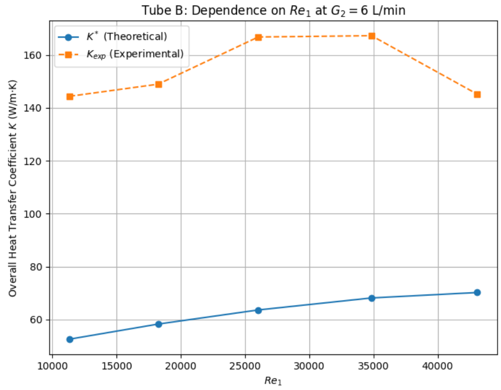


(d) $G_1 = 12$ L/min

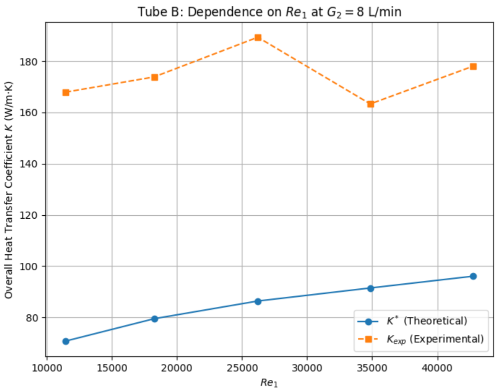
Figure 2: Tube B: Dependence of K on Re_2 at varying hot water flow rates G_1



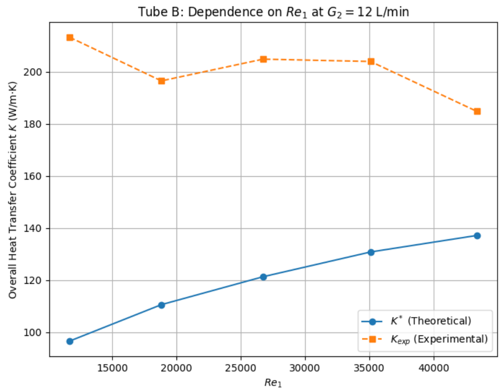
(a) $G_2 = 4$ L/min



(b) $G_2 = 6$ L/min



(c) $G_2 = 8$ L/min



(d) $G_2 = 12$ L/min

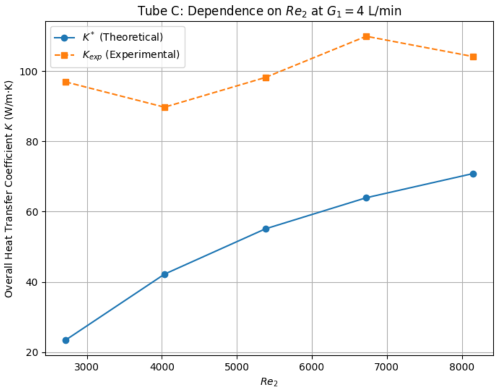
Figure 3: Tube B: Dependence of K on Re_1 at varying cold water flow rates G_2

Table 17: K stars and K experimental - Tube C

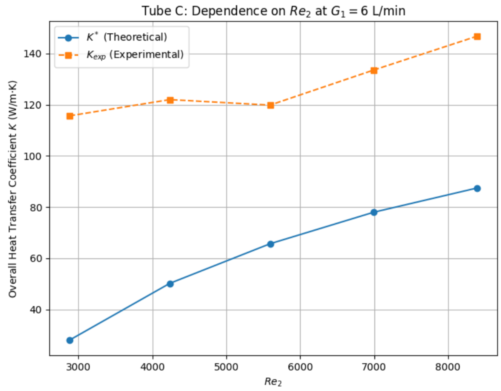
No.	$\boldsymbol{K^*}$ (W/m·K)	$\boldsymbol{K_{exp}}$ (W/m·K)	$\boldsymbol{Error}$ (%)
1	23.45	96.93	75.81
2	42.21	89.73	52.96
3	55.10	98.19	43.89
4	63.95	109.92	41.82
5	70.83	104.16	32.00
6	28.03	115.69	75.77
7	50.20	122.02	58.86
8	65.68	119.88	45.21
9	77.98	133.55	41.61
10	87.46	146.78	40.42
11	30.92	132.30	76.63
12	56.05	137.85	59.34
13	75.11	156.31	51.95
14	88.75	143.32	38.08

Table 17 – continued from previous page

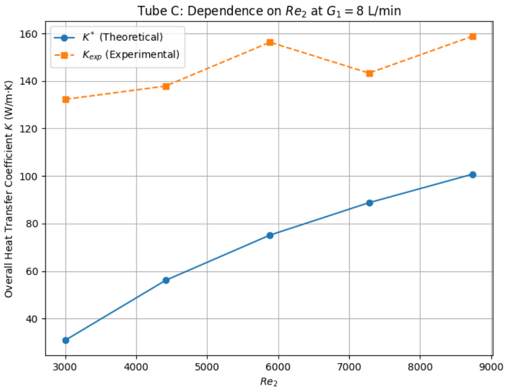
No.	K^* (W/m·K)	K_{exp} (W/m·K)	$Error$ (%)
15	100.74	158.82	36.57
16	33.42	137.40	75.67
17	61.04	153.75	60.30
18	81.52	161.30	49.46
19	98.32	169.17	41.88
20	110.97	170.55	34.93
21	34.77	153.70	77.38
22	64.44	153.57	58.04
23	86.45	174.70	50.51
24	104.72	178.85	41.45
25	119.82	195.48	38.70



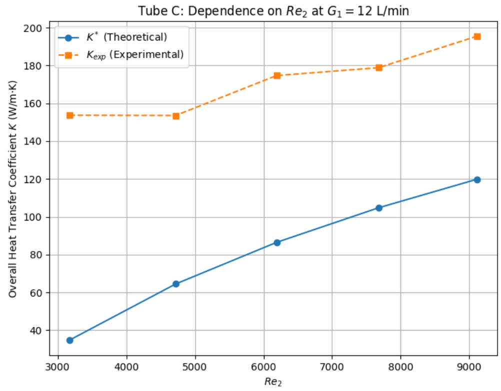
(a) $G_1 = 4$ L/min



(b) $G_1 = 6$ L/min

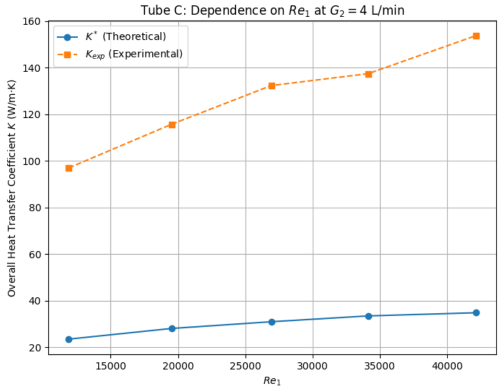


(c) $G_1 = 8$ L/min

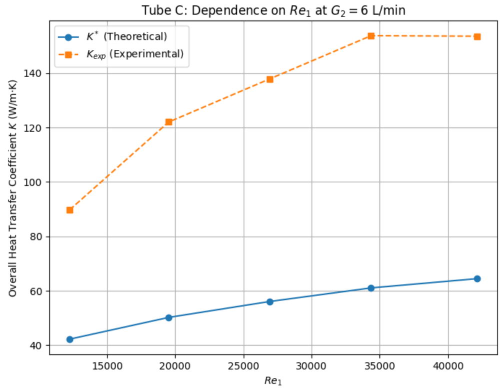


(d) $G_1 = 12$ L/min

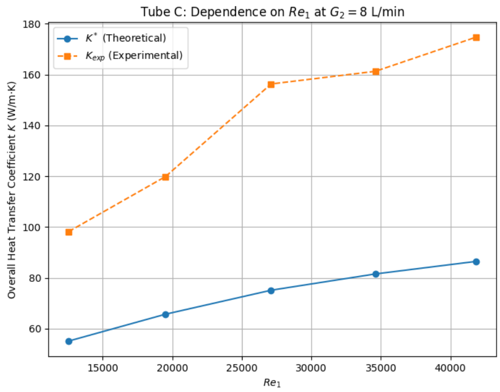
Figure 4: Tube C: Dependence of K on Re_2 at varying hot water flow rates G_1



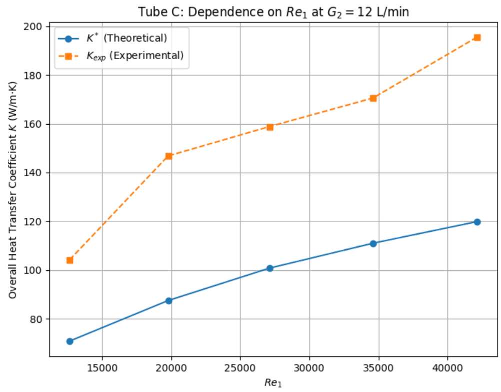
(a) $G_2 = 4$ L/min



(b) $G_2 = 6$ L/min



(c) $G_2 = 8$ L/min



(d) $G_2 = 12$ L/min

Figure 5: Tube C: Dependence of K on Re_1 at varying cold water flow rates G_2

Table 18: K dependence on flow mode Tube B

No.	$\boldsymbol{Re_2}$	$\boldsymbol{\alpha_1}$ (W/m^2·K)	$\boldsymbol{\alpha_2}$ (W/m^2·K)	$\boldsymbol{K^*}$ (W/m·K)
1	2882.1	5297.5	620.1	27.4
2	4367.8	5057.5	1374.7	52.6
3	5764.2	4968.1	2091.1	70.7
4	7205.2	4914.9	2815.6	85.1
5	8646.3	4870.1	3530.7	96.5
6	3005.2	6914.5	686.4	30.9
7	4507.8	6609.6	1455.5	58.3
8	6010.4	6466.0	2205.7	79.4
9	7433.5	6393.3	2935.2	96.3
10	8920.2	6344.2	3679.1	110.5
11	3171.2	8400.2	769.8	35.0
12	4680.3	8050.5	1549.1	63.6
13	6141.7	7881.8	2301.9	86.3
14	7594.1	7765.2	3043.8	105.0
15	9112.9	7690.3	3800.8	121.3

Table 18 – continued from previous page

No.	Re_2	α_1 (W/m^2·K)	α_2 (W/m^2·K)	K^* (W/m·K)
16	3251.1	9601.7	800.8	36.7
17	4836.0	9390.4	1630.9	68.1
18	6291.0	9114.7	2371.9	91.4
19	7863.8	8989.3	3153.3	112.4
20	9436.5	8933.4	3943.0	130.8
21	3306.7	10707.7	821.8	37.9
22	4918.0	10465.5	1655.3	70.2
23	6448.1	10276.7	2442.0	96.0
24	7993.5	10135.0	3221.7	118.0
25	9592.2	10022.1	4004.8	137.1

Table 19: K dependence on flow mode Tube C

No.	Re_2	α_1 (W/m^2·K)	α_2 (W/m^2·K)	K^* (W/m·K)
1	2715.5	3158.0	562.1	23.4
2	4034.5	2930.1	1253.1	42.2
3	5379.3	2873.5	1953.8	55.1
4	6724.2	2828.3	2637.4	64.0
5	8146.6	2802.5	3343.3	70.8
6	2882.1	4518.3	650.5	28.0
7	4236.5	4237.5	1372.2	50.2
8	5592.6	4094.4	2068.8	65.7
9	6990.7	4040.2	2785.7	78.0
10	8388.9	4000.1	3493.4	87.5
11	3005.2	5714.9	702.8	30.9
12	4413.5	5466.1	1459.9	56.0
13	5884.7	5327.3	2213.0	75.1
14	7279.7	5192.4	2911.3	88.8
15	8735.7	5149.0	3646.5	100.7
16	3120.2	6845.0	749.5	33.4
17	4606.2	6639.0	1542.0	61.0
18	6075.3	6452.8	2290.0	81.5
19	7594.1	6343.8	3045.8	98.3
20	8920.2	6253.1	3739.2	111.0
21	3171.2	7950.4	769.8	34.8
22	4718.3	7749.1	1587.8	64.4
23	6190.6	7536.2	2341.9	86.5
24	7677.1	7420.2	3094.2	104.7
25	9112.9	7337.8	3831.7	119.8

5 DISCUSSION

Question 1: Is heat loss significant? Why?

From calculation, it can be seen that the heat loss is significant, more detail, the heat loss in tube B is generally more than the one in tube C.

Causes:

- Heat loss is the result of heat transfer from the fluids to the environment, it may be because of a lack of good insulation.
- The heat is not only loss to the environment but also is 'trapped' in the metal wall. Also, copper has very high thermal-conductivity.
- The flow rates are not really stable during the experiment, due to the instability in pump work, which leads to unstable heat transfer.
- Tube B has more heat loss than tube C because the fluid streams in tube B contact each other in vertical flow leading to shorter heat transfer time in comparison with tube C. Beside that, fluid leaking on tube B is also a cause.
- Temperature values were displayed and observed subjectively leading to some cases that the inlet and outlet temperature were equal, which means there was no heat transfer, which is incorrect and does not reflect the nature of the heat exchanging process.

Question 2: Errors percentage and their causes? How to avoid them?

Errors percentage is significant. There are several causes:

- Flow rates of the hot and cold streams are not stable during the experiment. It is impossible to maintain a constant flow rate.
- The temperature changes continuously leading to errors in observing results.
- The hot stream is refluxed continuously leading to a non-constant hot stream inlet temperature.
- The higher the temperature, the greater the heat loss.
- Insulation is not good enough.
- There is fluid leaking during the experiment.
- Errors occur in flow meter and thermal-meter (system error).
- The experiment is performed once only, as the result, the repetition of the results is not confirmed.

- Students made mistakes in performing the experiment.
- Rounding number also contributes to the errors. The observed results are average numbers, which may leads to inaccurate calculation.

How to avoid errors:

- Supply a good insulation cover, repair leaks on the tubes.
- Perform the experiment in not-too-high temperature.
- Observe and note down the results quickly.
- Repeat the experiment more than once.
- Remove fouling inside tubes if possible.

Question 3: Compare experimental heat transfer coefficient K_l with overall heat transfer coefficient K_l^*.

- In this experiment, K_l is greater than K_l^* in most cases, which may be impossible. Because errors during the experiment are much greater than heat loss.
- The average wall temperature is estimated during calculation leading to incorrect Pr_{wall} in comparison with the experiment.
- The complex heat transfer had the student ignore some factor and only note down the main factors. For example, convectional heat transfer in tube B was ignored despite of the fact that convectional heat transfer plays an important role in how heat transfer is functioned.
- The difference between K_l and K_l^* is the result of significant heat loss, heat transfer from outside, fouling, ...During calculations, fouling was ignored since it is impossible to measure the fouling, which leads to noticeable differences between experimental and overall K.
- While K_l^* keeps its stability and slightly increases linearly, K_l fluctuates in the most cases.
- The experimental heat transfer coefficient of cold streams is lower than the overall due to the low heat transfer, because heat transfer area is small in comparison with flow rate, and there may be also fouling.
- The experiment heat transfer coefficient of hot stream is higher than the overall due to the heat is absorbed by the copper wall, more over, copper has higher thermal-conductivity than water.

In conclusion: The high thermal-conductivity copper wall which is covered by fouling is an important reason for heat loss, heat 'trapped' leading to huge difference between overall heat transfer coefficient and experimental one.

6 APPENDIX

6.1 Physical Parameters

The density ρ (kg/m^3), specific heat capacity C_p ($J/kg \cdot K$), thermal conductivity λ ($W/m \cdot K$), and dynamic viscosity μ ($N \cdot s/m^2$) are taken from standard property tables for water at the average fluid temperature.

The Prandtl number is calculated from these parameters using the formula:

$$Pr = \frac{\mu \cdot C_p}{\lambda} \tag{20}$$

6.2 Heat Q Calculation

Heat Q_1 of the hot stream transfers to the cold stream, the heat Q_2 that the cold stream receives is calculated as:

$$Q = G \cdot C \cdot (t_{in} - t_{out}) \tag{21}$$

Where:

- G: fluid flow rate, kg/s
- C: heat capacity of fluid, $J/kg \cdot K$
- t_{in}, t_{out}: inlet and outlet temperature of fluid, $°C$

Actually, there is heat loss, the heat transfer balance is established as:

$$\Delta Q = G_1 C_1 (t_{1in} - t_{1out}) - G_2 C_2 (t_{2out} - t_{2in}) \quad [W] \tag{22}$$

Where ΔQ is the heat loss.

6.3 Specific Geometric Dimension

In case of the fluid moves through a round section, specific geometric dimension is the diameter of the section. In case of the fluid moves through a non-round section, specific geometric dimension is determined as the equivalent diameter $d_{equivalent}$:

$$d_{eq} = \frac{4F}{\Pi} \tag{23}$$

Where:

- F: area of the section, m^2
- Π: wet circumference of the section, m

Tube B (Cross-flow):

- $l_1 = d_{1inner} = 0.014$ m
- $l_2 = d_{2inner} = 0.026$ m (equivalent diameter calculation based on annulus area and wetted perimeter).

6.4 Reynolds Number

Flow velocity w is determined from the volumetric flow rate V:

$$w = \frac{V}{F} \quad [m/s] \tag{24}$$

The Reynolds number (Re) is then calculated as:

$$Re = \frac{w \cdot l}{\nu} \tag{25}$$

6.5 Thermal Transmittance Coefficient α_1, α_2

The coefficient α is determined from the Nusselt number (Nu):

$$\alpha = \frac{Nu \cdot \lambda}{l} \tag{26}$$

Nusselt number is calculated in several formulas depending on flow mode in each tube:

- **Tube B (Hot stream):** $10^3 \leq Re < 2 \cdot 10^5$:

$$Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.4} \tag{27}$$

- **Tube B (Cold stream):** $5 < Re < 100$:

$$Nu = 0.5 \cdot Re^{0.5} \cdot Pr^{0.38} \cdot \left(\frac{Pr}{Pr_{wall}}\right)^{0.25} \tag{28}$$

- **Tube C:** $Re > 10^4$:

$$Nu = 0.021 \cdot Re^{0.8} \cdot Pr^{0.43} \cdot \left(\frac{Pr}{Pr_{wall}}\right)^{0.25} \cdot \epsilon_l \tag{29}$$

6.6 Overall Heat Transfer Coefficient K_l^*

From the value of α for each flow mode of each type of tube, choosing the thermal conductivity of the copper tube $\lambda_w = 385$ W/m.K, K_l^* is calculated as:

$$K_l^* = \frac{\pi}{\frac{1}{\alpha_1 d_1} + \frac{1}{2\lambda_w} \ln\left(\frac{d_2}{d_1}\right) + \frac{1}{\alpha_2 d_2}} \quad [W/m.K] \tag{30}$$

7 REFERENCES

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2. **Nguyen Bin and Partners:** *"Handbook Process and Equipment of Technology and Chemical"*, Vol. 1-2, Science-Technical Publisher.